Plasma proteomic associations with genetics and health in the UK Biobank

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Cohort: UK Biobank Pharma Proteomics Project (UKB-PPP)

N = 54306

Olink Ex

Study Design & Population

54306	cross-sectional biobank	plasma	separate
Sample Size	Design	Matrix	Multi-ancestry

Ancestry Breakdown

Ancestry Group	N
European	46327
South Asian	3810
African	2211
East Asian	895
Admixed American	285

Cohort(s)

UK Biobank Pharma Proteomics Project (UKB-PPP)

Ancestry-specific pQTLs: Yes | Count: 1163

Olink Platform & Proteomics QC

Olink Explore 3072 v1	2941	2730
Panel	Proteins Assayed	Proteins with ≥1 pQTL

NPX QC Parameters

Batch/Plate Correction	Olink internal batch normalization using bridge samples
Intensity Normalization	Quantile normalization applied per plate
LOD Handling	Proteins with >25% samples below LOD excluded; remaining below-LOD set to LOD/sqrt(2)
LOD Exclusion Threshold	25

Bridging & Multi-batch

Bridging Performed	Yes
Bridge Samples (N)	200
Bridging Reference	UKB-PPP pilot cohort

Genotyping & GWAS Methodology

Genotyping

Genotyping Array	UK Biobank Axiom Array
Imputation Reference	HRC r1.1 + UK10K
Imputation R^2	0.3
MAF Threshold	0.001

⚠️ *MAF Note: MAF ≥ 0.001 appropriate given N > 50,000; yields ~1 expected minor allele carrier per 500 individuals at minimum*

GWAS Methods

Statistical Model	Linear regression via REGENIE v3.2.1
Genome-wide Threshold	$5×10^{-8}$
Protein Bonferroni Correction	Yes
cis Threshold	$1.7×10^{-11}$ ($5×10^{-8}$ / 2941 proteins)
trans Threshold	$5×10^{-8}$
cis Window	1000 kb (TSS)

Covariates

age, sex, BMI, top 20 genetic PCs, Olink plate, sample dilution factor, frozen sample indicator, UKB assessment centre

Post-GWAS Analyses

Fine-Mapping

Performed:	Yes
Method:	SuSiE
Credible Sets:	14287
PIP Threshold:	0.9

Colocalization

Performed:	Yes
Method:	coloc v5
PP4 Threshold:	0.8
Traits Tested:	GTEx v8 eQTLs, GWAS Catalog disease loci, eQTLGen blood eQTLs

Epitope Artifacts

Assessed:	Yes
Method:	Comparison with SomaScan v4.1 pQTLs; SNPs within 10bp of antibody epitope flagged
Flagged:	89

Colocalization Key Findings

1,547 protein-disease colocalizations identified; notable examples include IL6R with inflammatory bowel disease, PCSK9 with LDL cholesterol, and APOE with Alzheimer's disease

Trans-pQTL Hotspots

ABO (187 proteins); HLA region (143 proteins); FUT2 (61 proteins); APOE (38 proteins)

Mendelian Randomization & Data Access

MR Performed: Yes

Proteins (exposures)	IL6R, PCSK9, GDF15, VEGFA, TNF
Outcomes tested	Coronary artery disease, Type 2 diabetes, Alzheimer's disease, BMI, LDL cholesterol
Methods used	IVW, MR-Egger, weighted median, MR-PRESSO
F-statistics reported	Yes

Key MR Findings

PCSK9 → LDL: OR 0.71 per SD ($p=4.2\times10^{-12}$); IL6R → CAD: OR 0.87 per SD ($p=0.003$); GDF15 → BMI: $\beta=-0.12$ ($p=6.1\times10^{-8}$)

✓ Summary statistics PUBLICLY AVAILABLE

GWAS Catalog: GCST90293000–GCST90295940; open access via UK Biobank AMS portal